

Business Insights 360

Project Overview

AtliQ Hardware — Analytics Transformation Initiative

Document Type	Project Overview
Project Name	Business Insights 360
Organization	AtliQ Hardware
Domain	Manufacturing & Hardware Distribution
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1. Executive Summary

AtliQ Hardware operates across multiple regions, selling computer hardware and peripherals to retail chains, e-commerce platforms, and direct clients. As the company scaled globally, its reliance on Excel-based reporting created a significant gap between the volume of data being generated and the quality of insights reaching decision-makers.

This project delivered an end-to-end analytics solution: structured data exploration via SQL, followed by a multi-page Power BI dashboard covering Finance, Sales, Marketing, Supply Chain, and Executive reporting. The outcome is a centralized, self-serve reporting environment where each department can track its key metrics, identify trends, and act on data rather than intuition.

The dashboard is live on Power BI Service with automatic data refresh via gateway local Excel files.

2. Business Background

AtliQ Hardware manufactures and distributes hardware products globally across multiple channels — retail, distributor, and direct. The company serves a diverse customer base, ranging from large retailers to regional resellers, and tracks performance across divisions such as PC, Peripherals & Accessories (P&A), and Networking & Storage (N&S).

Historically, reporting was handled through Excel workbooks maintained by individual teams. Each department tracked its own numbers, which led to inconsistent metrics, version conflicts, and reporting delays. As the business grew into new international markets, the data volume outpaced what Excel could handle reliably, and leadership began making decisions without a consolidated view of performance.

The analytics transformation was initiated by the top management to address this gap — moving from fragmented, manual reporting to a governed, automated BI environment.

3. Business Problem

The core problem was not a lack of data — it was the inability to use that data effectively. Specific issues included:

- Finance had no real-time view of P&L metrics across markets, making it difficult to monitor Net Sales, Gross Margin, and Net Profit against targets.
- Sales teams lacked visibility into customer-level and product-level profitability, often spending time building reports instead of acting on them.

- Marketing could not reliably measure how operational spending affected gross margin or net profit.
- Supply Chain had no consistent way to track forecast accuracy or flag inventory risk (overstock or stockout) before it became a problem.
- Leadership had no unified view — market share, revenue contribution, and top-performing products or customers were scattered across separate reports.

A failed market entry in the Latin American region — attributed partly to decisions made without adequate data — accelerated the push for a proper BI solution.

4. Project Objectives

The project had two phases with clear deliverables:

Phase 1 — SQL Analysis

- Extract and validate key business metrics from the MySQL database.
- Write queries and stored procedures to produce on-demand reports for product owners.
- Derive P&L metrics, top/bottom product and customer rankings, and supply chain forecast accuracy.

Phase 2 — Power BI Dashboard

- Connect Power BI to MySQL and Excel; clean and model the data using Power Query.
- Build a Snowflake schema data model and create DAX measures for all KPIs.
- Deliver five dashboard views — Finance, Sales, Marketing, Supply Chain, and Executive.
- Publish to Power BI Service with gateway-enabled auto-refresh.
- Incorporate stakeholder feedback and complete user acceptance testing (UAT).

5. Project Scope

In Scope	Out of Scope
Finance View: P&L statement, KPIs, Net Sales trend	Predictive or machine learning models
Sales View: Customer & product performance, unit economics	Real-time streaming data
Marketing View: Net Profit analysis, performance matrix	CRM or ERP integration beyond source data provided
Supply Chain View: Forecast accuracy, risk analysis	Mobile-specific dashboard design
Executive View: Market share, revenue by channel/division	Data entry or write-back functionality
SQL queries & stored procedures for key reports	Automated email alerts or subscriptions
Data pipeline via Power Query (ETL)	Financial forecasting or budgeting modules
Snowflake schema data modelling in Power BI	
Published to Power BI Service with auto-refresh	

6. Stakeholders

Stakeholder	Role	Primary Interest
Top Management / CEO	Executive Sponsor	Overall company performance, market share, revenue by division
Finance Team	Primary User	P&L metrics, Gross Margin, Net Profit vs targets
Sales Team	Primary User	Customer and product performance, discount impact
Marketing Team	Primary User	Operational expense impact on Net Profit and Gross Margin
Supply Chain Team	Primary User	Forecast accuracy, inventory risk, Net Error trends
Analytics / BI Team	Project Owner	Data pipeline, dashboard quality, UAT, deployment
Product Owners	Report Consumer	SQL-generated on-demand reports via stored procedures

7. Technology Stack

Layer	Tool / Technology
Data Storage	MySQL (relational database, Snowflake schema ERD)
Data Analysis (SQL)	MySQL Workbench — queries, stored procedures, P&L reports
Data Source (Flat File)	Microsoft Excel — supplementary data files
ETL / Data Preparation	Power Query (M language) within Power BI
Data Modelling	Power BI — Snowflake schema relationships
Calculations & Metrics	DAX — calculated columns and measures
Visualization	Microsoft Power BI Desktop
Publishing & Sharing	Power BI Service with gateway for auto-refresh
Project Management	Project charter, stakeholder mapping, Kanban board

8. High-Level Solution

The solution follows a standard BI pipeline from source to report:

Source data — transactional records, customer and product master data, and targets — was loaded into MySQL. The schema was designed in Snowflake format, with fact and dimension tables linked via an ERD. SQL queries and stored procedures were written at this layer to generate reports on demand.

Power BI connects to both MySQL and Excel using native connectors. Power Query handles all transformation — renaming columns, handling nulls, deriving calculated fields, and merging tables before they reach the data model. The data model replicates the Snowflake schema structure, and all KPIs are built as DAX measures rather than calculated columns where possible, to improve performance.

The final output is a five-page dashboard published to Power BI Service. A data gateway keeps the reports current by refreshing automatically when the source data changes. Stakeholder feedback captured during UAT was used to add an Executive View and resolve data quality issues identified during testing.

9. Expected Business Benefits

- Faster decisions — department heads can access current KPIs without waiting for manual reports.
- Consistent numbers — a single data model ensures all teams are working from the same metrics.
- Proactive supply chain management — forecast accuracy and risk flags allow inventory issues to be caught early rather than after the fact.
- Clearer P&L visibility — finance can track Gross Margin and Net Profit across markets, products, and time periods in one place.
- Reduced reporting effort — stored procedures and automated refresh eliminate the manual work of pulling and formatting data.
- Executive alignment — leadership has a single view of market share, revenue trends, and top performers without needing to consolidate multiple reports.